

Correction

Thermodynamique

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Chapitre 1

Exercice 1

1) $M = 18 \text{ g/mol}$
 $\rho = 1000 \text{ kg/m}^3$

$$V = 1 \text{ m}^3$$

$$n = \frac{\rho}{M} = \frac{1}{18} = 55555 \text{ mol/m}^3$$

$$N = n \cdot N_A = 3,3 \cdot 10^{28} \text{ particules/m}^3$$

2) Gaz d_g :

$$PV = nRT \Rightarrow n = \frac{PV}{RT} = 0,03 \text{ mol} \cdot \text{L}^{-1}$$

pour $V = 1 \text{ L}$:

$$x = n \cdot N_A = 1,8 \cdot 10^{22}$$

$$d_p = \frac{x}{V} = 1,8 \cdot 10^{25} \text{ particules} \cdot \text{m}^{-3}$$

$$\frac{d_g}{d_p} = 0,00054$$

3): $V_m = \frac{V}{N} = d^3$

Liquide: $V_m = 2,9 \cdot 10^{-29} \text{ m}^3$
 $= 29 \text{ \AA}^3$

$$d_p = \underline{3,1 \text{ \AA}}$$

gaz: $V_m = \frac{1}{d_g} = 5,6 \cdot 10^{-24} \text{ m}^3 = 5,6 \cdot 10^6 \text{ \AA}^3$

$$d_g = \underline{38 \text{ \AA}}$$

Exercice 2

1) $PV = nRT$

$$n = \frac{PV}{RT} = 5507 \text{ mol}$$

$$\alpha = 3,32 \cdot 10^{23} \text{ atomes}$$

~~$m = 44 \text{ g mol}^{-1}$~~

$$V = 185 \text{ m}^3$$

$$T = 273 \text{ K}$$

$$P = 10^5 \text{ Pa}$$

2) Énergie interne : $U = \bar{\epsilon}_c^{\text{int}} + \bar{\epsilon}_p^{\text{int}} = \frac{1}{2} m \langle v^2 \rangle$

$$= \frac{3}{2} R_B T$$

$$\Rightarrow m \langle v^2 \rangle = 3 R_B T$$

$$\Rightarrow \langle v^2 \rangle = \frac{3 R_B T}{m}$$

$$m = \frac{M}{N_A}$$

$$\Rightarrow v = \sqrt{\frac{3RT}{M}} = \underline{1305 \text{ m.s}^{-1}}$$

3) $\bar{\epsilon}_c^{\text{moy}} = \langle \bar{\epsilon}_c \rangle = \frac{1}{2} m_{\text{Tot}} \langle v^2 \rangle$

$$= N \cdot \frac{1}{2} m \langle v^2 \rangle$$

$$= N \cdot \frac{3}{2} R_B T$$

$$= m \cdot N_A R_B \frac{3}{2} T$$

$$= m R \frac{3}{2} T$$

$$= 19000 \text{ RJ}$$

Énergie interne $U = 19000 \text{ RJ}$

$$= \frac{3}{2} nRT$$

4) $m = 10000 \text{ kg}$

$$\bar{\epsilon}_p = mgz$$

$$z = \frac{\bar{\epsilon}_p}{mg}$$

$$z = \frac{19 \cdot 10^6}{10000 \cdot 9.8} = \underline{190 \text{ m}}$$

Exercice 3

2

Extensive : proportionnelle à la taille du système
Intensive : indépendante " " "

$P = \text{intensive}$
 $m, n = \text{intensive}$

Variable	Extensive	Intensive	common
V	✓		
$1/V$			✓
V/N		✓	
V^2/N	✓		✓
V^2			✓
P/V			✓
PV	✓		
T		✓	
$1/T$		✓	
mV			✓

$$\frac{\text{ext}}{\text{ext}} = \text{int}$$

$$\text{ext} \cdot \text{int} = \text{ext}$$

$$N = \text{ext}$$

$$m = \text{ext}$$

Exercice 4

$$V = 60 \text{ m}^3$$

$$R = 200 \, \Omega$$

$$U = 220 \text{ V}$$

Système isolé.

(sans radiateur)

$$P = UI$$

$$= \frac{U^2}{R} W$$

$$T_0 = 16^\circ \text{C} = 289 \text{ K}$$

$$P_0 = 1,013 \cdot 10^5 \text{ Pa}$$

1) $dW = P dt$

Sur un temps Δt , $W_e = P \Delta t = \frac{U^2}{R} \Delta t$
 ~~$= 568 \text{ J}$~~

2) W_e est intégralement convertie en chaleur qui est elle-même cédée au système.

(isolé)

$$Q_{\text{ext}} = W_e = \Delta U$$

$$\Delta U = C_V \Delta T$$

$$= \frac{5}{2} n R \Delta T$$

Donc pour chauffer le système de ΔT il faut lui fournir ΔU

$$n = \frac{P_0 V_0}{RT_0} = 2530 \text{ mol}$$

$$\Rightarrow \Delta U = Q_{\text{ext}} = \frac{U^2}{R} \Delta t$$

$$= \frac{5}{2} n R \Delta T$$

$$\Delta t = 870 \text{ s} \approx 14,5 \text{ min}$$

Exercice 5

$n = 20 \text{ mol}$ de GPM
état initial. $P_0 = 10^5 \text{ Pa}$

$$T_0 = 400 \text{ K}$$

état final $P_1 = 1,2 \cdot 10^5 \text{ Pa}$

$$T_1 = 415 \text{ K}$$

1) $V_0 = 0,665 \text{ m}^3$
 $V_1 = 0,575 \text{ m}^3$

2) Izobare : $P = \text{cte}$
Izochore : $V = \text{cte}$

3) $P_a = P_0$ } $T_A = 346 \text{ K}$ 4) $P_B = P_1$ } $T_B = 480 \text{ K}$
 $V_a = V_1$ } $V_B = V_0$ }

5) $\delta W_A = -P_{\text{ext}} dV$
 $= \delta W_A^{(1)} + \delta W_A^{(2)} \rightarrow = 0 \text{ (izochore)}$
 $\Rightarrow \int P_0 dV$ car $P_{\text{ext}} = P_0 = \text{cte}$ tout au long de
 $W_A = -P_0 \Delta V$ la transformation.
 $= \underline{3000 \text{ J}}$

6) $\delta W_B = -P_{\text{ext}} dV$
 $= \delta W_B^{(1)} + \delta W_B^{(2)}$
 $W_B = -P_1 \Delta V$
 $= \underline{10800 \text{ J}}$

7) Premier principe

$$\Delta U = W + Q_{\text{ext}}$$

$$Q_A = \cancel{W} - \Delta U = \Delta U - W$$

$$= \frac{3}{2} n R (T_1 - T_0) - 9000$$

$$= \underline{-5260 \text{ J}}$$

$$Q_B = \Delta U - W$$

$$= \underline{-7059 \text{ J}}$$

8) a) Les amènent du même état initial
→ final
≠ énergétiquement

b) IP cède de la chaleur

c) La temp ↑ mais on donne plus de travail
qu'il ne perd de chaleur.

Exercice 6

1) $V_0 = 0,0166 \text{ m}^3$

$$V_1 = 0,0249 \text{ m}^3$$

2) Durant la détente $A \rightarrow B$

$$W_{A \rightarrow B} = -P_1 \Delta V = \text{---} 1245 \text{ J} \text{ --- } 830 \text{ J}$$

$$\begin{aligned} \delta W_{A \rightarrow B} &= -P_{\text{ext}} dV \\ &= -P_1 dV \end{aligned}$$

3) $\delta W_{B \rightarrow A} = -P_{\text{ext}} dV$

P_{ext} : équilibre mécanique $\Rightarrow P_{\text{ext}} = P_{\text{int}}$

$$P_{\text{ext}} = \frac{nRT}{V}$$

Car, $B \rightarrow A$ est isotherme

A tout moment, $T = T_0$

$$\delta W_{B \rightarrow A} = - \frac{nRT_0}{V} dV$$

$$W_{B \rightarrow A} = \int_{V_1}^{V_0} nRT_0 \frac{1}{V} dV$$

$$= nRT_0 \ln\left(\frac{V_1}{V_0}\right)$$

$$= \underline{1021 \text{ J}} > 0$$

4) $P_1 V_1 = nRT_1 = nRT_0$

$$W_{B \rightarrow A} = P_1 V_1 \ln\left(\frac{V_1}{V_0}\right)$$

$$\begin{aligned}
 \underline{5)} \quad W_T &= W_A + W_B \\
 &= -P_1(V_1 - V_0) + P_1 V_1 P_m \left(\frac{V_1}{V_0} \right) \\
 &= 181 \text{ J} > 0
 \end{aligned}$$

Le système gagne du travail mécanique.

$$\begin{aligned}
 \underline{6)} \quad x &= \frac{V_0}{V_1} & W_T &= P_1 V_1 (-1 + x + P_m \left(\frac{1}{x} \right)) \\
 & & &= P_1 V_1 (-1 + x - P_m(x))
 \end{aligned}$$

$$\underline{7)} \quad 0 < P_1 < P_0 \quad x = \frac{V_0}{V_1}, \quad V_0 = \frac{nRT_0}{P_0}$$

$$V_1 = \frac{3nRT_1}{2P_1}$$

$$\Rightarrow x = \frac{V_0}{V_1} = \frac{P_1}{P_0}$$

$$0 < x < 1$$

$$g'(x) = P_1 V_1 \left(1 - \frac{1}{x} \right)$$

$$0 < x < 1$$

$$\frac{1}{x} > 1$$

$$-\frac{1}{x} < -1$$

$$1 - \frac{1}{x} < 0$$

$$\text{donc } g'(x) < 0$$

$$x = 0$$

$$1$$

$$g'(x)$$

$$g(x)$$

$$\text{donc } g(x) \geq 0 \quad \forall x$$

$$\underline{8)} \quad W + Q = 0 \quad \Rightarrow Q = +840 \text{ J}$$

Exercise 7

$$C \rightarrow A : P_c = 5 \text{ bar}, V_c = V_A$$

$$A \rightarrow B : T_A = 300 \text{ K}, P_A = 1 \text{ bar}$$

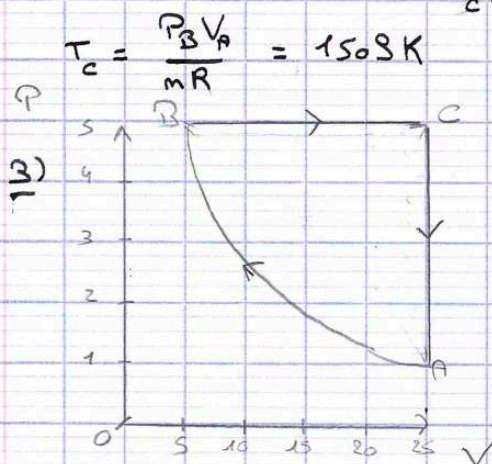
$$B \rightarrow C : T_B = T_A, P_B = 5 \text{ bar}$$

- 1) Isotherme réversible
 Isobare ou (quasi) statique
 Isochore

2) $PV = nRT$
 $V_A = 25 \text{ L}$

$$V_B = \frac{nRT_A}{P_B} = 5 \text{ L}$$

$$V_C = V_A = 25 \text{ L}$$



4) AB: $\delta W_{A \rightarrow B} = - \frac{nRT_A}{V} dV$

$$W_{A \rightarrow B} = -nRT_A \ln\left(\frac{V_B}{V_A}\right) = -4015 \text{ J}$$

BC: $W_{B \rightarrow C} = -10000 \text{ J}$

CA: $W_{C \rightarrow A} = 0 \text{ J} (\Delta V = 0)$

5) $Q = \Delta U - W$ où $\Delta U = \frac{3}{2} nRT$

AB: $Q = -4015 \text{ J} (\Delta T = 0)$

BC: $Q = \frac{3}{2} nR(T_C - T_B) - W$
 $= 2604 \text{ J}$

CA: $Q = \frac{3}{2} nR(T_C - T_A)$
 $= -15015 \text{ J}$

Bilan total:

$$\Delta U = \sum \Delta U_{mn} = 0$$

Travail

$$W = \sum W_{mn} = -5986 \text{ J}$$

Chaleur

$$Q_T = \sum Q_{mn} = 5986 \text{ J}$$

Exercice 8

1) Transformation brusque non statig

2) Variation du volume du pressostat

$$V_{\text{initial}} = V_{\text{tot}} - V_0 - V_R$$

$$V_{\text{final}} = V_T - V_R$$

$$\text{donc } \Delta V = V_0$$

3) Le système applique une pression de P_0 sur le pressostat.

4) Travail reçu par le système :

$$\delta W = -P_{\text{ext}} dV$$

$$= -P_{\text{pressostat}} dV$$

$$= -P_{\text{pressostat}} dV$$

$$= -P_0 dV$$

$$W = -P_0 \Delta V = -P_0 (V_R - V_0) \quad \text{or on ne peut pas résoudre}$$

$$\delta W' = -P_{\text{ext}} dV$$

$$= -P_0 dV$$

$$W' = -P_0 \Delta V$$

$$= -P_0 V_0$$

$$\text{donc } W = -W' \\ = P_0 V_0$$

5) $\Delta U = \frac{3}{2} nR (T_R - T_0) = nRT_0$

$$\Rightarrow T_R = \frac{5}{3} T_0 = 500 \text{ K}$$

Exercice 9

Comment savoir T_8

(1) Système fermé, isolé, pas de chaleur échangée

$$m_1 \text{ à } T_1 \rightarrow T_8$$

$$m_2 \text{ à } T_2 \rightarrow T_8$$

$$\text{calorimètre } T_1 \rightarrow T_8$$

$$H_1 = H_1 + H_2 + H_c$$

$$H_8 = H_1' + H_2' + H_c'$$

$$\begin{aligned} \Delta H = H_8 - H_1 &= (H_1' - H_1) + (H_2' - H_2) + (H_c' - H_c) = 0 \\ &= \Delta H_1 + \Delta H_2 + \Delta H_c \end{aligned}$$

$$\Delta H_1 = C_p \Delta T$$

$$\Delta H = m_1 C_{eau}^m (T_8 - T_1) + m_2 C_{eau}^m (T_8 - T_2) + C (T_8 - T_1) = 0$$

$$C (T_8 - T_1) = \frac{m_1 C_{eau}^m (T_8 - T_1) + m_2 C_{eau}^m (T_8 - T_2)}{T_8 - T_1}$$

Exercice 10

$$(1) U_1 = U_1 + U_2 + U_c = m_1 C_{eau}^m T_2 + m_2 C_{eau}^m T_2 + 0$$

$$U_8 = (m_1 + m_2) C_{eau}^m T_{eq}$$

$$\text{Système isolé : } \Delta U = 0 \Leftrightarrow U_8 = U_1$$

$$(m_1 + m_2) C_{eau}^m T_{eq} = (m_1 T_1 + m_2 T_2) C_{eau}^m$$

$$T_{eq} = 51,8^\circ \text{C}$$

$$(2) U_1 = (m_1 C_{eau}^m + C) T_1 + m_2 C_{eau}^m T_2$$

$$U_8 = [(m_1 + m_2) C_{eau}^m + C] T_{eq}'$$

$$\Delta U = 0$$

$$(m_1 + m_2) C_{eau}^m T_{eq} + C T_{eq} = (m_1 T_1 + m_2 T_2) C_{eau}^m + C T_1$$

$$C (T_{eq} - T_1) = C_{eau}^m [(m_1 T_1 + m_2 T_2) - (m_1 + m_2) T_{eq}]$$

$$C = 115,5 \text{ J.K}^{-1}$$

III - Entropie et Second principe

Exercice 11

(1) Ferme (impermeable) et isolé (cloisons indeformable / adiabatique)

(2) 1^{er} principe : $dU = \delta W + \delta Q$
 $= 0 + 0 = dU$

$$\Delta U = 0$$

(3) $U_1 = U_{11} + U_{12}$

$$= \frac{3}{2} m R T_1 + \frac{3}{2} m R T_2$$

$$= \frac{3}{2} m R (T_1 + T_2)$$

ou $\Delta U = 0$

$$\Leftrightarrow U_8 = U_1$$

$$\Leftrightarrow T_8 = \frac{T_1 + T_2}{2}$$

$$U_8 = U_{81} + U_{82} = 3 m R T_8$$

$N = \text{nbr particule}$

(4) $S = R_B \ln(\Omega)$

$$\Omega = \Omega_{em} \Omega_{pos}$$

$$\Omega_{em} = cte U^{3N/2} \quad (U \propto T)$$

$$= cte T^{3N/2}$$

$$\Omega_{pos} = \frac{N!}{N_0! (N_0 - N)!}$$

$$\Delta S = S_8 - S_1$$

$$S_1 = S_{11} + S_{12} = R_B \ln(cte T_1^{3N/2} C_{N_0}^N) + R_B \ln(cte T_2^{3N/2} C_{N_0}^N)$$

$$= R_B \ln(cte \times T_1^{3N/2} T_2^{3N/2} (C_{N_0}^N)^2)$$

$$S_g = S_{g1} + S_{g2} \\ = k_B \ln (c k_B T_g^{3N/2} C_{N_0}^N) + k_B \ln (c k_B T_g^{3N/2} C_{N_0}^N)$$

$$S_g = k_B \ln [c k_B (T_g^{3N/2})^2 (C_{N_0}^N)^2] \\ = k_B \ln [c k_B T_g^{3N} (C_{N_0}^N)^2]$$

$$\Delta S = S_g - S_i = k_B \ln [c k_B T_g^{3N} (C_{N_0}^N)^2] - k_B \ln [c k_B T_1^{3N/2} T_2^{3N/2} (C_{N_0}^N)^2] \\ = k_B \ln \left[\frac{c k_B T_g^{3N} (C_{N_0}^N)^2}{c k_B (T_1 T_2)^{3N/2} (C_{N_0}^N)^2} \right] = k_B \ln \left(\frac{T_g^{3N}}{(T_1 T_2)^{3N/2}} \right) \\ = \frac{3N}{2} k_B \ln (T_g^2 / T_1 T_2)$$

Signe de ΔS

$$T_g^2 = \frac{T_1^2 + T_2^2 + 2T_1 T_2}{4}$$

$$\frac{T_g^2}{T_1 T_2} = \frac{T_1^2 + T_2^2 + 2T_1 T_2}{4T_1 T_2} > 1$$

irréversible

$$\Delta S \neq 0.$$

$$(T_g - T_1)^2 = T_1^2 + T_2^2 - 2T_1 T_2 > 0$$

$$T_1^2 + T_2^2 > 2T_1 T_2$$

$$\Leftrightarrow T_1^2 + T_2^2 + 2T_1 T_2 > 4T_1 T_2$$

$$\Rightarrow N > D$$

$$(s) \quad dU = -p dV + T dS + \mu dW \quad \text{fermé : } dW = 0$$

$$U = \frac{3}{2} nRT \quad \left. \begin{array}{l} \text{impossible car} \\ \text{transformation} \end{array} \right\}$$

$$dU = \frac{3}{2} nR dT \quad \left. \begin{array}{l} \text{irréversible} \end{array} \right\}$$

$$dV = 0$$

$$dU = T dS$$

$$dS = \frac{dU}{T}$$

car entropie est une variable d'état donc elle ne dépend que de l'état initial et final. On fait donc comme si elle était réversible

$$\text{et } dS_1 = \frac{3}{2} m R \frac{dT}{T}$$

$$\Delta S_1 = \int_{T_1}^{T_8} \frac{3}{2} m R \frac{dT}{T}$$

$$\Delta S_1 = \frac{3}{2} m R \ln\left(\frac{T_8}{T_1}\right)$$

$$\Delta S_{\text{totale}} = \frac{3}{2} m R \ln\left(\frac{T_8^2}{T_1 T_2}\right)$$

$$(6) \Delta S_2 = \frac{3}{2} m R \ln\left(\frac{T_8}{T_2}\right)$$

$$T_2 > T_1, \quad T_8 = \frac{T_1 + T_2}{2}$$

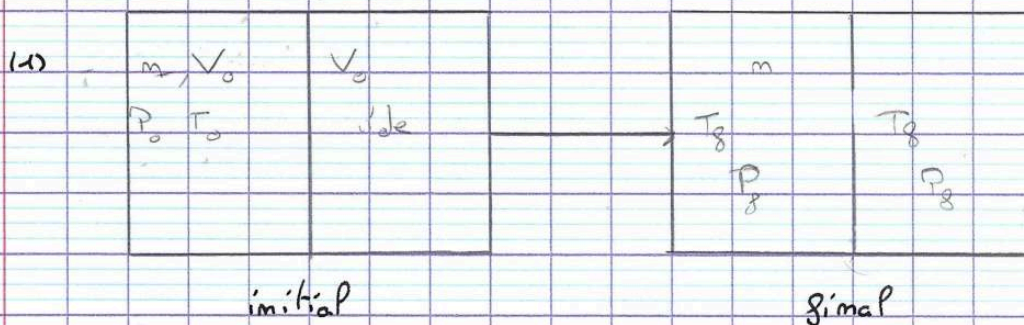
$$T_2 > T_8 \Rightarrow \ln\left(\frac{T_8}{T_2}\right) < 0 \Rightarrow \Delta S_2 < 0$$

de même

$$\Delta S_2 = \int_{T_2}^{T_8} \frac{3}{2} m R \frac{dT}{T}$$

$$= \frac{3}{2} m R \ln\left(\frac{T_8}{T_2}\right)$$

Exercice 12 - Détente de Joule - Gay-Lussac Thermo



$$V_g = 2V_0$$

(2) $\Delta U = \delta W + \delta Q$ hors enceinte adiabatique
 $\Rightarrow dU = 0$ incompressible

(3) $\Delta U = \frac{3}{2} m R \Delta T = 0$
 $\Rightarrow T_g = T_0$

$$P_g V_g = m R T_g = m R T_0$$

$$2 P_g V_0 = m R T_0$$

$$P_g = P_0 / 2$$

(4) Identité thermodynamique : $dU = -p dV + T dS + \mu dN$
 $\Rightarrow T dS = P dV$
 $dS = \frac{P}{T} dV$
 $= -P dV + T dS$
 $= 0$

Si la transformation était réversible, on aurait

P définie à tout moment. $P = \frac{m R T_0}{V}$

$$\Rightarrow dS = \frac{m R T_0}{V} \frac{dV}{T_0} = m R \frac{dV}{V}$$

$$\Rightarrow \Delta S = \int_{V_0}^{2V_0} m R \frac{dV}{V} = m R \ln(2)$$

(5) ΔS par Boltzmann

$$S = k_B \ln(\Omega)$$

$$\Delta S = S_g - S_i$$

$$S_i = k_B \ln \left(C' T_0^{3N/2} \frac{C_{N_0}^N}{C_{N_0}^N} \right)$$

$$S_g = k_B \ln \left(C' T_0^{3N/2} \frac{C_{2N_0}^N}{C_{N_0}^N} \right)$$

$$\Omega_e = C U^{3N/2}$$

$$\Omega_p = \frac{C_{N_0}^N}{N! (N_0 - N)!} = \frac{N_0!}{N! (N_0 - N)!}$$

$$\begin{aligned}
 \Delta S &= S_g - S_i \\
 &= R_B \left[\ln \left(C_{T_o}^{3N/2} C_{w_o}^N \right) - \ln \left(C_{T_o}'^{3N/2} C_{w_o}'^N \right) \right] \times R \times \\
 &= R_B \ln \left[\frac{C_{w_o}^N}{C_{w_o}'^N} \right] = R_B \ln (2^N) = N R_B \ln (2) = m N_A R_B \ln (2) \\
 &= m R \ln (2)
 \end{aligned}$$

Exercise 13

$$(1) \quad dU = -P dV + T ds + \mu dN$$

$$T ds = dU + P dV$$

$$ds = \frac{dU}{T} + \frac{P}{T} dV$$

$$(2) \text{ GPM: } U = \frac{3}{2} m R T = C_V T = m C_V T$$

$$dU = C_V dT \quad \text{et} \quad PV = m R T \Rightarrow P = \frac{m R T}{V}$$

$$\Rightarrow ds = \frac{C_V}{T} dT + \frac{1}{T} \frac{m R T}{V} dV = C_V \frac{dT}{T} + \frac{m R}{V} dV$$

$$(3) \quad s(x, y) = A dx + B dy = ds$$

$$\text{soi: } \frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}$$

$$\begin{aligned}
 \text{ici: } A &= \frac{C_V}{T}, \quad dT = dx \\
 B &= \frac{m R}{V}, \quad dV = dy
 \end{aligned}
 \left. \vphantom{\begin{aligned} A &= \frac{C_V}{T} \\ B &= \frac{m R}{V} \end{aligned}} \right\} \begin{array}{l} \text{d'où } ds \text{ est une diff} \\ \text{totale} \end{array}$$

$$ds = C_V \frac{dT}{T} + \frac{m R}{V} dV$$

$$(4) \quad A = \frac{C_V}{T} = \frac{\partial S}{\partial T} \quad B = \frac{m R}{V} = \frac{\partial S}{\partial V}$$

$$\Rightarrow \frac{\partial S}{\partial T} = \frac{C_V}{T} \Rightarrow S(T, V) = C_V \ln(T) + K(V)$$

$$\Rightarrow \frac{\partial S}{\partial V} = \frac{\partial K(V)}{\partial V} = \frac{m R}{V} \Rightarrow K(V) = m R \ln(V) + K$$

$$\Rightarrow S(T, V) = C_V \rho_m T + m R \rho_m V + K$$

$$C_V = \frac{3}{2} m R = \frac{3}{2} \frac{N}{N_A} N_A R_3$$

$$= \frac{3}{2} N R_3 \quad \text{et} \quad m R = N R_3$$

$$\begin{aligned} \Rightarrow S &= \frac{3}{2} N R_3 \rho_m T + N R_3 \rho_m V + C \\ &= R_3 \left(\frac{3}{2} N \rho_m T + N \rho_m V \right) + C \\ &= R_3 \left(\rho_m T + \rho_m V + C \right) \\ &= R_3 \left(C \rho_m T + \rho_m V \right) \end{aligned}$$

$$\begin{aligned} 5) \quad S &= R_3 \rho_m (\Omega) \\ &= R_3 \rho_m (C T^{3/2} V_0^N) \end{aligned}$$

$$\begin{aligned} \Omega &= \Omega_0 e^{\Omega_0} = C V^{3/2} C^{N_0} \\ &= C^N T^{3/2} C^{N_0} \\ &= C_T^{(3)} T^{3/2} \left(\frac{N_0}{N} \right)^N \\ &= C_T^{(4)} T^{3/2} \frac{N_0^N}{N^N} \\ &= C T^{3/2} V_0^N \end{aligned}$$

Exercice 14

$$(1) \quad P_0, T_0, V_0 = \frac{m R T_0}{P_0}$$

$$\begin{aligned} \text{à l'état final} \quad P_8 &= P_0 + P_m \\ &= P_0 + \frac{m g}{S} \end{aligned}$$

$$\begin{aligned} T_8 &= T_0 \\ V_8 &= \frac{m R T_8}{P_8} = \frac{m R T_0}{P_0 + \frac{m g}{S}} \end{aligned}$$

(2) en

$$\begin{aligned} (3) \quad \delta W_1 &= -P_{\text{ext}} dV \\ &= -P_{\text{int}} dV \\ &= -\frac{m R T_0}{V} dV \end{aligned}$$

$$\begin{aligned} W_1 &= -m R T_0 \rho_m \left(\frac{V_8}{V_0} \right) = m R T_0 \rho_m \left(\frac{V_0}{V_8} \right) \\ \frac{V_8}{V_0} &= \frac{P_0}{P_8} = \frac{1,6 m g}{5 P_0} \\ \Rightarrow W_1 &= m R T_0 \rho_m \left(1 + \frac{m g}{5 P_0} \right) \end{aligned}$$

(4)



$$(5) \quad dU_x = C_v dT = 0 \quad \text{car } T = \text{cte} = T_0$$

$$= \delta W_x + \delta Q_x$$

$$\delta Q_x = -\delta W_x$$

$$Q_x = -W_x = -mRT_0 \ln\left(1 + \frac{M_0}{SP_0}\right)$$

Chaleur cédée

$$(6) \quad \Delta S = \underbrace{dU}_0 = -p dV + T dS + \underbrace{\mu dN}_0$$

$$T dS = p dV$$

$$dS = \frac{p}{T} dV = \frac{mRT}{T_0 V} dV = \frac{mR}{V} dV$$

$$\Delta S = \int_{V_0}^{V_g} dS = \int_{V_0}^{V_g} mR \frac{dV}{V} = mR \ln\left(\frac{V_g}{V_0}\right) = mR \ln\left(\frac{P_0}{P_g}\right)$$

$$= -mR \ln\left(1 + \frac{M_0}{SP_0}\right)$$

$$(7) \quad \Delta S^e : \quad dS^e = \frac{\delta Q_{\text{resu}}}{T_{\text{ext}}} = \frac{\delta Q_x}{T_0}$$

$$\Delta S^e = \int_1^8 dS^e = \int_0^1 \frac{\delta Q_x}{T_0} = \frac{Q_x}{T_0}$$

$$= \frac{-mRT_0}{T_0} \ln\left(1 + \frac{M_0}{SP_0}\right) = -mR \ln\left(1 + \frac{M_0}{SP_0}\right)$$

$$\Delta S_{\text{sys}} = \Delta S^e + \Delta S^i$$

$$\Delta S^i = \Delta S_{\text{sys}} - \Delta S^e = 0 \quad \text{car } \Delta S_{\text{sys}} = \Delta S^e$$

$$(13) \quad du = -pdv + Tds = 0$$

$$Tds = pdv$$

$$ds = \frac{pdv}{T} = \frac{pdv}{T_0}$$

$$\Delta S = -mR \ln \left(1 + \frac{m_0}{SP_0} \right)$$

OR p non définie mais
 S est une variable d'état donc
 ΔS ne dépend pas du type de
transformation donc ΔS sera le
même que pour la transformation
1 car les états initiaux et
finaux sont les mêmes

$$(14) \quad \cancel{\Delta S} \quad dS^e = \frac{\delta Q_{\text{recu}}}{T_{\text{ext}}} = \frac{\delta Q_e}{T_0}$$

$$\Delta S^e = \frac{Q_e}{T_0} = -\frac{m_0}{ST_0} V_0$$

$$\Delta S_{\text{univ}}^c = \Delta S_{\text{sys}} - \Delta S^e$$

$$= -mR \ln \left(1 + \frac{m_0}{SP_0} \right) + \frac{m_0}{ST_0} V_0 > 0$$

$$\Delta S_{\text{univ}} = \Delta S^c$$

La transformation est irréversible

$$\begin{aligned}
 \Delta S^{\text{universe}} &= \Delta S_{\text{sys}} + \Delta S_{\text{ext}} \\
 &= \Delta S^c + \Delta S^e - \Delta S^e \\
 &= \Delta S^c \\
 &= 0 \Rightarrow \text{transformation reversible}
 \end{aligned}$$

(8) état initial : P_0, V_0, T_0

état final : $P_g = P_0 + \frac{m_g}{S}$

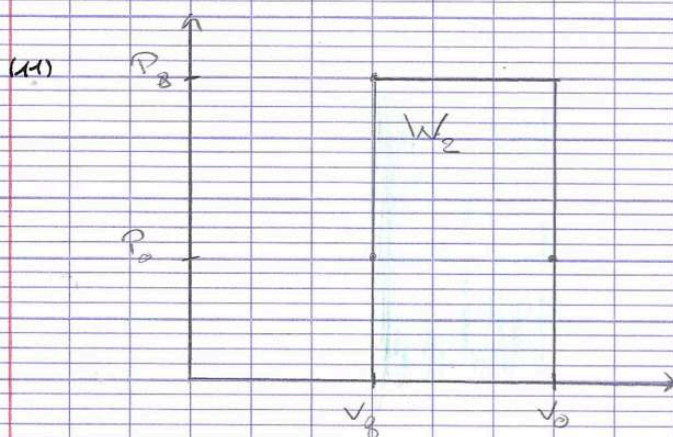
$$\begin{aligned}
 T_g &= T_0 \quad \text{car la transformation est très rapide.} \\
 V_g &= \frac{nRT_0}{P_g}
 \end{aligned}$$

(9) Non $\Rightarrow$ brutale

$$\begin{aligned}
 (10) \quad \delta W_2 &= -P_{\text{ext}} dV = -P_g dV \\
 W_2 &= \int_{V_0}^{V_g} -P_g dV = -P_g \int_{V_0}^{V_g} dV = -P_g (V_g - V_0)
 \end{aligned}$$

$$\begin{aligned}
 W_2 &= -P_g V_g + P_g V_0 \\
 &= -P_g V_g + \left(P_0 + \frac{m_g}{S}\right) V_0 = -P_g V_g + P_0 V_0 + \frac{m_g V_0}{S} = \frac{m_g V_0}{S}
 \end{aligned}$$

$$P_0 V_0 = nRT_0 = nRT_g = P_g V_g$$



$$\begin{aligned}
 (12) \quad dU_2 &= \delta W_2 + \delta Q_2 = C_V dT \\
 &= 0
 \end{aligned}$$

$$\Rightarrow \Delta U_2 = C_V \Delta T = 0 \quad \text{car } T_g = T_0$$

$$\Delta U_2 = W_2 + Q_2 \Rightarrow Q_2 = -W_2 = -\frac{m_g V_0}{S}$$